

Δv Sensitivity of Plane-Change Timing in Hohmann Transfers

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Abstract

This paper investigates how the timing of an orbital plane-change maneuver affects total Δv during simplified Hohmann transfers. Using a two-body impulsive-burn model, total Δv was compared for plane changes performed at periapsis and apoapsis. A sample 400 km to 800 km transfer with a 47° inclination change showed that apoapsis timing reduced total Δv by approximately 346 m/s. The analysis was then extended by varying inclination change and final orbit altitude to examine how the Δv savings change across different transfer cases. Results show that apoapsis timing required less total Δv for all outward-transfer cases tested under this separate-burn model, with the advantage increasing when the velocity difference between periapsis and apoapsis becomes larger.

Research Question

The goal of this project is to quantify how performing a plane change at periapsis versus apoapsis affects the total Δv required during a simplified Hohmann transfer. Specifically, this paper asks: under what transfer conditions does apoapsis plane-change timing provide the greatest Δv savings?

Assumptions

This model assumes:

- The initial and final orbits are circular,
- The spacecraft is modeled in a two-body Earth-spacecraft system,
- Burns are impulsive (each maneuver occurs instantaneously),
- Atmospheric drag is neglected,
- Orbital perturbations, including J_2 , are neglected,
- Rendezvous phasing and launch-site constraints are not included.
- All transfers considered are outward transfers, meaning the final orbit radius is greater than the initial orbit radius.
- The Hohmann transfer is used as an altitude-transfer geometry between circular orbits.
- Plane changes are modeled as separate impulsive maneuvers.
- In the periapsis case, the plane change is performed immediately after transfer-orbit insertion.
- In the apoapsis case, the plane change is performed immediately before circularization.
- When two impulses occur at the same location, they are treated as separate burns rather than as one combined burn.
- For the orbital-pair comparison, $\Delta\theta = |i_2 - i_1|$, assuming equal RAANs and an appropriate common line of nodes.
- Earth's radius was taken as $R_E = 6371$ km, with orbital radius calculated using $r = R_E + h$.

Theory and Method

Circular Orbital Velocity

$$v_c = \sqrt{\frac{\mu}{r}}$$

Used to calculate the velocity of a spacecraft in a circular orbit, where μ is Earth's gravitational parameter and r is the orbital radius [1]. Here,

$$\mu = 3.986 \times 10^{14} \frac{\text{m}^3}{\text{s}^2}$$

Transfer Orbit Velocity

$$v_t = \sqrt{\mu \left(\frac{2}{r} - \frac{1}{a} \right)}$$

The vis-viva equation was used to calculate the spacecraft's velocity at different points along the transfer ellipse [1]. Here, a is the semi-major axis of the transfer orbit:

$$a = \frac{r_1 + r_2}{2}$$

Hohmann Transfer Δv

$$\Delta v_1 = |v_{t,p} - v_{c1}|$$

$$\Delta v_2 = |v_{c2} - v_{t,a}|$$

These terms represent the two burns required to enter and exit the transfer orbit [2]. Here, v_{c1} and v_{c2} are the circular velocities of the initial and final orbits, while $v_{t,p}$ and $v_{t,a}$ are the transfer-orbit velocities at periapsis and apoapsis.

Plane Change Δv

$$\Delta v_{\text{plane}} = 2v \sin \left(\frac{\Delta \theta}{2} \right)$$

The plane-change equation was evaluated at two possible locations [3]:

- periapsis, where $v = v_{t,p}$
- apoapsis, where $v = v_{t,a}$

Total Δv

$$\Delta v_{\text{total}} = \Delta v_1 + \Delta v_{\text{plane}} + \Delta v_2$$

The total Δv was calculated separately for the periapsis and apoapsis plane-change cases. In this simplified model, the altitude change follows Hohmann-transfer geometry, while the plane change is treated as a separate impulse. In the periapsis case, the plane change occurs after transfer-orbit insertion; in the apoapsis case, it occurs before circularization.

Δv Savings Relationship

Because Δv_1 and Δv_2 are the same in both cases, they cancel when the difference between the two total Δv values is calculated:

$$\Delta v_p = (\Delta v_1 + \Delta v_2) + 2v_{t,p} \sin \left(\frac{\Delta \theta}{2} \right)$$

$$\Delta v_a = (\Delta v_1 + \Delta v_2) + 2v_{t,a} \sin \left(\frac{\Delta \theta}{2} \right)$$

Therefore

$$\Delta v_{\text{savings}} = \Delta v_p - \Delta v_a = 2(v_{t,p} - v_{t,a}) \sin \left(\frac{\Delta \theta}{2} \right)$$

For the outward transfers considered in this model, $v_{t,p}$ is greater than $v_{t,a}$. Therefore, the savings are positive for any nonzero plane-change angle. This relationship also shows that the savings increase when either the plane-change angle or the velocity difference between periapsis and apoapsis increases.

Example Calculation

A transfer from a 400 km orbit to an 800 km orbit with a 47° inclination change was used as a sample case. The orbital radii are $r_1 = 6.771 \times 10^6$ m and $r_2 = 7.171 \times 10^6$ m, giving circular velocities of $v_{c1} = 7672.6$ m/s and $v_{c2} = 7455.5$ m/s. The transfer velocities are $v_{t,p} = 7781.9$ m/s and $v_{t,a} = 7347.8$ m/s.

The two Hohmann-transfer burns are:

$$\Delta v_1 = |7781.9 - 7672.6| = 109.3 \text{ m/s}$$

$$\Delta v_2 = |7455.5 - 7347.8| = 107.7 \text{ m/s}$$

The plane-change costs at periapsis and apoapsis are:

$$\Delta v_{\text{plane},p} = 2(7781.9)\sin(47^\circ/2) = 6206.0 \text{ m/s}$$

$$\Delta v_{\text{plane},a} = 2(7347.8)\sin(47^\circ/2) = 5859.9 \text{ m/s}$$

The resulting total Δv values are:

$$\Delta v_p = 109.3 + 6206.0 + 107.7 = 6423.0 \text{ m/s}$$

$$\Delta v_a = 109.3 + 5859.9 + 107.7 = 6076.9 \text{ m/s}$$

Therefore, performing the plane change at apoapsis saves approximately:

$$\Delta v_p - \Delta v_a = 346.1 \text{ m/s}$$

Primary Results: Effect of Inclination Change

After evaluating the 47° example case, the inclination change was varied from 5° to 65° while keeping the initial and final orbit altitudes fixed at 400 km and 800 km. This sweep was used to examine how the Δv difference between periapsis and apoapsis plane-change timing changes as the inclination angle increases.

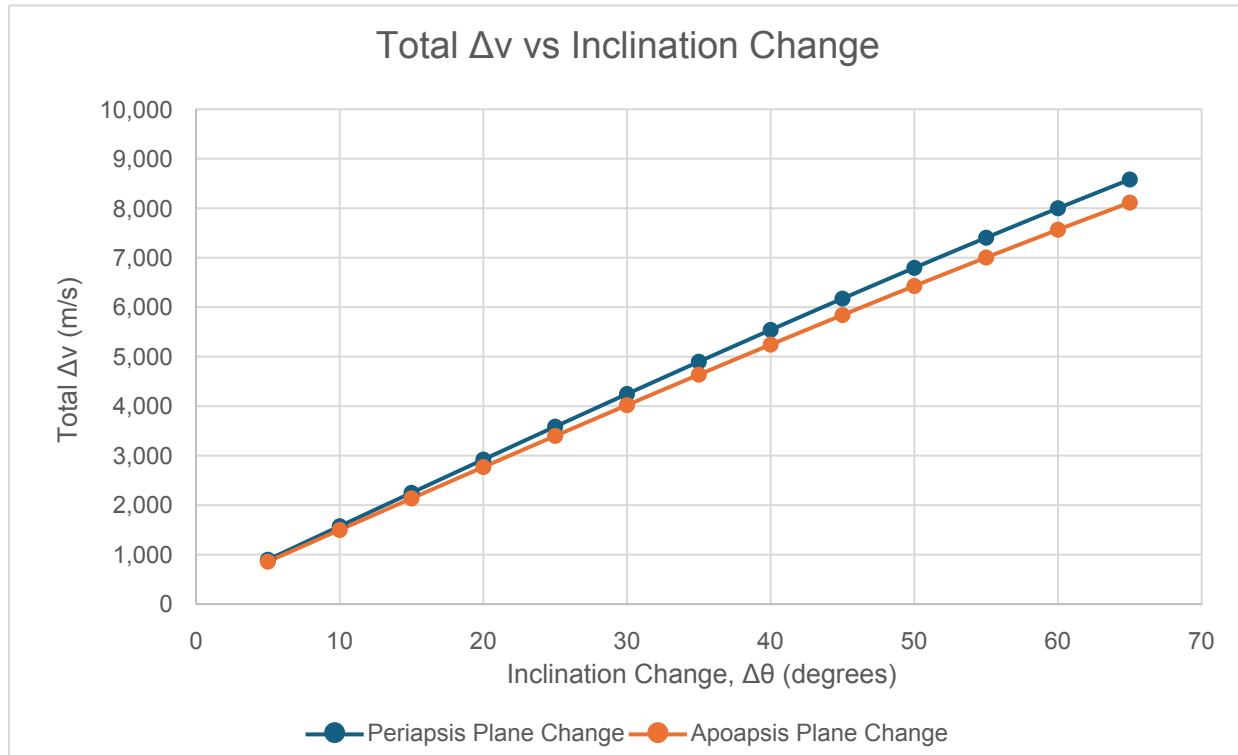


Figure 1. Total Δv as a function of inclination change for a transfer from 400 km to 800 km altitude. The two curves compare cases where the plane change is performed at periapsis or apoapsis.

Figure 1 shows that performing the plane change at apoapsis requires less total Δv than performing it at periapsis across all tested inclination angles. The difference becomes larger as the inclination change increases because plane-change Δv depends directly on local orbital velocity. For example, at a 47° inclination change, the apoapsis case requires about 346 m/s less Δv than the periapsis case.

Multi-Target Analysis

To test whether the apoapsis advantage holds across different orbital cases, several representative low-Earth-orbit targets were selected. These include ISS-like, polar, sun-synchronous-like, and higher-altitude LEO orbits. Pairwise transfers were then evaluated to compare how inclination change and orbital radius separation affect Δv savings.

Table 1: Orbital Targets and Parameters

Target	Type	Altitude (km)	Inclination (°)	Orbital Radius (m)
T_1	ISS-like	400	51.6	6.77×10^6
T_2	LEO	500	53	6.87×10^6
T_3	LEO	600	70	6.97×10^6
T_4	SSO like	600	97.6	6.97×10^6
T_5	SSO like	800	98.6	7.17×10^6
T_6	Polar	800	90	7.17×10^6
T_7	High LEO	1200	63	7.57×10^6
T_8	High LEO	1400	74	7.77×10^6

Orbital targets used in Table 2, ordered from lowest to highest altitude.

Table 2: Orbital Pair Δv Comparison

Pair	$\Delta\theta$ (°)	Total Δv_p (m/s)	Total Δv_a (m/s)	$\Delta v_{Difference}$ (m/s)
$T_1 \rightarrow T_2$	1.4	244.2	241.5	2.74
$T_7 \rightarrow T_8$	11	1493.9	1457.9	36.0
$T_3 \rightarrow T_6$	20	2750.9	2677.1	73.8
$T_1 \rightarrow T_8$	22.4	3591.4	3194.9	396.5
$T_5 \rightarrow T_8$	24.6	3533.1	3283.0	250.1
$T_3 \rightarrow T_5$	28.6	3868.0	3763.1	104.9
$T_4 \rightarrow T_7$	34.6	4894.9	4531.2	363.7
$T_2 \rightarrow T_4$	44.6	5856.0	5772.7	83.2

Here, Δv_p is the total maneuver Δv when the plane change is performed at periapsis, Δv_a is the total maneuver Δv when the plane change is performed at apoapsis, and the Δv difference is $\Delta v_p - \Delta v_a$. Differences were calculated from unrounded values

The results show that apoapsis timing remains more efficient across all tested target pairs. However, the Δv savings do not depend only on inclination change, but they also depend on the difference between the transfer velocity at periapsis and apoapsis. This explains why some transfers with moderate inclination changes can show larger savings than transfers with larger inclination changes but smaller radius differences.

Extended Analysis: Effect of Target Orbit Altitude

To isolate the effect of target orbit altitude, the initial orbit altitude was fixed at 400 km, and the inclination change was fixed at 47°. The final orbit altitude was then varied from 800 km to 6400 km. For each case, the total Δv was calculated for a plane change performed at periapsis and at apoapsis, and the difference between the two cases was recorded.

The Δv savings were calculated as:

$$\Delta v_{\text{savings}} = \Delta v_p - \Delta v_a$$

where Δv_p is the total Δv when the plane change is performed at periapsis, and Δv_a is the total Δv when the plane change is performed at apoapsis.

Table 3: Target Orbit Altitude Sweep

Altitude cases used for the target orbit altitude sweep. The initial orbit altitude is fixed at 400 km, and the inclination change is fixed at 47°. The total Δv values are rounded to the nearest 10 m/s; savings were calculated from unrounded values

Case	Final Altitude (km)	Δv_p (m/s)	Δv_a (m/s)	$\Delta v_p - \Delta v_a$ (m/s)
1	800	6420	6080	346
2	1000	6570	6060	509
3	1400	6840	6020	814
4	2000	7200	5970	1230
5	2800	7640	5920	1720
6	3800	8100	5860	2240
7	5000	8570	5800	2770
8	6400	9030	5740	3290

The results show that the Δv savings increase as the final orbit altitude increases. Over the tested range, the total Δv for the periapsis plane-change case increases from about 6.42 km/s to 9.03 km/s, while the total Δv for the apoapsis plane-change case decreases slightly from about 6.08 km/s to 5.74 km/s. This occurs because, in the apoapsis case, the reduction in plane-change cost at the slower and higher apoapsis more than offsets the additional Hohmann-transfer cost.

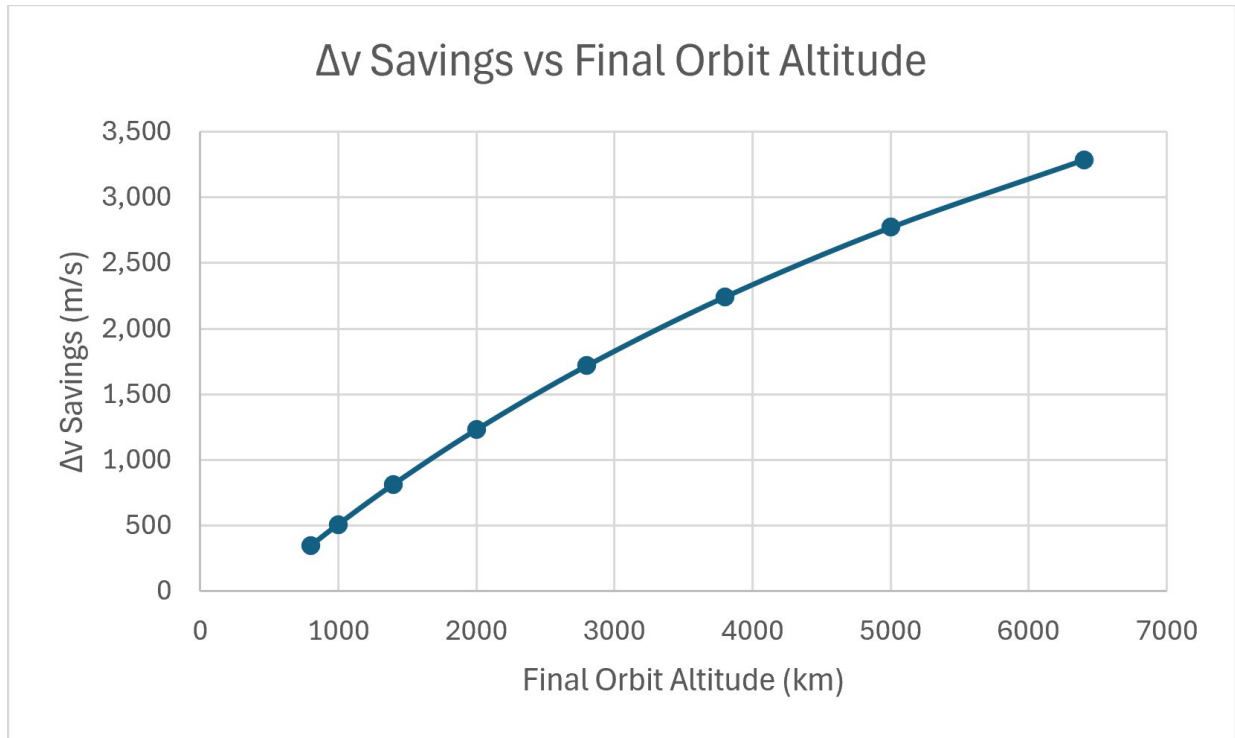


Figure 2: Δv savings from apoapsis plane-change timing as final orbit altitude increases. The initial orbit altitude is fixed at 400 km, and the inclination change is fixed at 47° .

Figure 2 shows that the advantage of apoapsis plane-change timing becomes larger as the final orbit altitude increases. This is because the spacecraft moves more slowly at higher apoapsis distances, reducing the cost of the plane change.

Limitations

This analysis is meant to isolate the effect of plane-change timing, so several real mission factors were left out. The model assumes circular orbits, impulsive burns, and a simplified two-body Earth-spacecraft system. Because of this, the results should be treated as an idealized Δv comparison rather than a complete mission design.

The main limitations are:

- Atmospheric drag was neglected
- J_2 perturbations were neglected
- Burns were treated as instantaneous
- Rendezvous phasing was not included
- Launch site constraints were not included
- The plane change was treated separately from the Hohmann transfer burns
- Combined burn-plane-change maneuvers were not modeled

Conclusion

This analysis showed that, for the outward-transfer cases tested under the separate-burn model, performing the plane change at apoapsis required less total Δv than performing it at periapsis. In the 400 km to 800 km example with a 47° inclination change, apoapsis timing reduced the total Δv by approximately 346 m/s. The inclination sweep showed that the savings increase as the plane-change angle becomes larger, while the target altitude sweep showed that apoapsis timing becomes more favorable as the final orbit altitude increases. Although the model neglects several real-world factors, the results demonstrate how maneuver timing can significantly affect Δv requirements in orbital transfers.

References

1. Massachusetts Institute of Technology. “Lecture L16 – Central Force Motion: Orbits.” *16.07 Dynamics*, MIT OpenCourseWare, Fall 2009.
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Use of AI Tools

AI tools were used to review the paper and help draft or revise some explanations, as well as to suggest improvements to grammar, clarity, and structure. The author evaluated these suggestions, completed and checked the calculations in Excel, created the graphs, and made the final decisions about the paper.